

Qualifying the Measurement Instrument

Phase 0 of the Phase 1 replacement programme — and five occasions on which a measurement artefact was read as a result about the method

August 2026

Abstract

Before approaches **B** (auction-dynamics MBO) and **C** (capacity-constrained geodesic Lloyd) can be compared against the PGD control on perimeter, the things that would do the comparing must themselves be shown to work: a *balanced-assignment solver* shared by all three approaches, and a single *evaluation harness*. This report records that qualification. Four results. (1) The harness reproduces the validated $N=100$ deliverable's Phase 2 perimeter **bitwise** — 185.2546144718457 — from a completely different entry path, which simultaneously establishes that hardening an arm's labels costs it nothing and that Phase 2 is bit-deterministic on this host. (2) The shared solver needed *no new algorithm*: the conclusion that an auction or Sinkhorn solver was required came from a fixture modelling a state the arms occupy for one outer iteration. (3) Under the final protocol (three meshes $\times$ two arms $\times$ three seeds) the solver passes **18/18**, and the gate is shown to discriminate — the pre-0a un-normalized solver is *indistinguishable from doing nothing*. (4) Measured at genuinely equal footing the arms sit at **parity**, $\pm 10\%$, with the incumbent, not clear of it; two earlier versions of this claim were wrong in the arms' favour. The methodological content is larger than the technical content: **five** times during this work a defect in the measurement was read as a result about the method, and the report documents what caught each one.

Provenance

Status	measured
Date	2026-08-16 to 2026-08-18
Surface	torus $T(R=1.0, r=0.6)$ throughout
Fixtures	$V=47,488$ (224×212) at $N=300$; $V=114,144$ (348×328) at $N=100$ and at $N=300$ (the production configuration). Seeds 0, 1, 2.
Control	<code>results/run_20260709_081548_surftorus_npart100_..._lam5.1_seed84172851</code> , the validated $N=100$ deliverable: Phase 2 best perimeter 185.2546144718457 at iteration 20 of 20, campaign <code>ipopt.btol0.001_lbfgs30_hess_bestiter_partial</code> , worst cell 0.7786%, 0 dead / 0 weak / 0 fragmented.
$N=300$ baselines	<code>run_20260806_123326</code> ($V=47,488$, Phase 2 best 323.319247087254, plateaued at iterate 18 of 19) and <code>run_20260808_191030</code> ($V=114,144$, 322.96218780465847, <i>censored</i> — best is its final iterate). These are <i>one</i> experiment: the second resumed from the first (<code>warm_start_path</code>). Both were copied into this checkout on 2026-08-18 (12.6 GB, verified identical).
Incumbent	approach A's own dual stage on its own log densities, normalized, at the same 2000-iteration budget the arms are scored with: 1.3517% at $V=47,488$, 0.5991% at $V=114,144$, both $N=300$.
Scripts	<code>testing/test_balanced_assignment_solver.py</code> (gates 1–4); <code>testing/test_arm_harness.py</code> (0b acceptance); <code>testing/test_phase0d_shakedown.py</code> (0d, ~ 34 min); <code>docs/experiments/07-phase0-shared-harness/make_figures.py</code> .
Code	<code>src/partition/balanced_readout.py</code> (<code>solve_dual_offsets</code> , <code>assignment_margin_scale</code>); <code>src/partition/arm_harness.py</code> (<code>assignment_quality_bar</code> , <code>A_PARITY_NORMALIZED</code>).
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1, h5py 3.16.0; macOS 15.3.1 arm64 (Mac mini).
Anchors	0d reproduces 185.2546144718457 bitwise ; final gate 2 is 18/18 in 2 h 39 m; the null solver scores 16.588% at the production pin against a 0.841% bar.

1 Where B and C come from

Neither arm is invented here; both are established constructions applied to this surface and this pipeline. This section states the provenance explicitly, because an earlier draft of this report carried none.

B — volume-constrained threshold dynamics. Threshold dynamics originates with Merriman, Bence and Osher [14]: alternate short-time diffusion of the phase indicators with a pointwise reassignment, and the interfaces move by (approximate) mean curvature. Esedoğlu and Otto [7] put the multiphase case on a variational footing, exhibiting a Lyapunov functional that Γ -converges to a weighted perimeter — the same limit our own energy targets [3, 15]. The specific variant we would build is Jacobs, Merkurjev and Esedoğlu's *auction dynamics* [10], which imposes exact volume constraints by replacing the pointwise threshold with a *balanced assignment*, solved by Bertsekas' auction algorithm [2]. That paper is the direct antecedent: it is volume-constrained multiphase MBO, published for this problem class.

Two caveats the literature itself supplies. Connectivity is *not* guaranteed per step — stray components shrink under curvature flow but are not excluded — and on graphs and coarse meshes threshold dynamics is known to *freeze* when the diffusion length falls to the discretization scale [16]. Both shaped the design: the first sets the validity grading, the second fixes $\tau = (4h)^2$ rather than h^2 .

C — capacity-constrained centroidal power diagrams. Lloyd’s algorithm [12] iterates “assign to nearest site, move site to centroid”. Adding a prescribed-capacity constraint turns the assignment into a semi-discrete optimal transport problem whose solution is a *power diagram* with per-site weights — the Minkowski-type result of Aurenhammer, Hoffmann and Aronov [1]. Mérigot [13] and Kitagawa, Mérigot and Thibert [11] give the modern solvers for those weights; de Goes *et al.* [6] use exactly this construction for blue-noise sampling. On a curved surface the Euclidean distance is replaced by a geodesic one, for which the heat method [4] is the standard alternative to the graph-Dijkstra approximation used in our fixtures.

C minimizes a *quantization* energy, not perimeter. It lands near the perimeter optimum only because both optima are asymptotically hexagonal in the plane — Hales for perimeter [9], Fejes Tóth and Gershgorin [8] for quantization. **Both are flat-plane results**, and this torus has Gaussian curvature of both signs, so neither transfers; how far C lands from optimal is precisely what remains to be measured.

The shared solver, and what was nearly rebuilt. All three approaches need the same primitive, and it is the semi-discrete OT dual [1, 13]: find per-cell offsets ψ such that $\arg \max_k [s_{ik} + \psi_k]$ has equal masses. The shipped implementation is subgradient ascent on that dual. Section 4 records that this report’s programme nearly replaced it with an auction [2] or an entropic (Sinkhorn) solver [5] — both standard, either defensible — on evidence that turned out to be a fixture artefact. Sinkhorn was implemented and measured; it *lost* to the incumbent on five of six score matrices.

2 Why Phase 0 exists

The Phase 1 relaxation costs **22.0–79.3 h** for a completed $N=300$ solution and its winner-take-all readout is not a valid partition, so it is repaired afterwards. That does not scale. Approaches B and C would produce a valid partition at every step in minutes. Deciding between them requires measuring *perimeter after Phase 2*, and both — plus the shipped approach A — need one shared primitive: *balanced assignment given arbitrary per-vertex scores*, which exists inside the balanced readout as `solve_dual_offsets`.

Phase 0 makes that primitive usable on foreign score scales (0a), builds one evaluation harness (0b), adopts the $N=300$ baselines (0c), and proves the harness reproduces a known answer (0d). **It measures nothing about perimeter**; every question about whether B or C is any good remains open.

3 0d — the harness reproduces a known answer, bitwise

The strongest single result. The $N=100$ control’s own winner-take-all labels were hardened to one-hot densities, written in the Phase 1 solution schema, and pushed through the entire harness under the control’s own campaign `refinement.yaml`:

reference	185.2546144718457	
harness	185.2546144718457	(iterate 20 of 20)
relative difference	0.000000%	bitwise: true wall 2023 s

Independently verified: the scratch campaign holds 20 fresh iteration files, its `refinement.yaml` names the one-hot arm file as `base_solution`, and the *entire* 20-iterate trajectory is bitwise equal to the control campaign computed five weeks earlier, including the 14,883-entry `lambda_parameters` array.

Three consequences. **(a)** The apples-to-apples question is settled exactly: an arm’s hard labels

and PGD’s continuous field produce identical Phase 2 initial states, because `ContourAnalyzer` reads densities only through `argmax` and every variable point starts at $\lambda=0.5$. (b) Phase 2 is bit-deterministic on this host, previously an unverified load-bearing assumption. (c) The control is itself *censored* — its best iterate is its last — so 185.2546144718457 is a *bound*.

4 0a — no new solver was needed

`solve_dual_offsets` is subgradient ascent on the semi-discrete OT dual. It is *correct* for any additive score matrix but its step schedule was calibrated to the log-density scale. Normalizing scores by a robust assignment-margin scale, and raising the step (`normalized_eta0` 0.5 $\rightarrow$ 2.0; measured 3.654 $\rightarrow$ 2.209 $\rightarrow$ **1.672** $\rightarrow$ 2.248% for $\eta_0 = 0.5, 1, 2, 4$) and the budget (400 $\rightarrow$ 2000) is the whole fix. Approach A’s path is untouched and byte-identical.

4.1 The conclusion that was withdrawn

The first version of gate 2 scored a *cold start* — balance a raw geometric guess in one shot — and B failed at 45.35%. This was written up as structural: the shipped solver is subgradient ascent tuned for a small correction, B and C need to rebalance ~ 230 of 300 cells, therefore build a Bertsekas auction or an entropic solver. **All of that was wrong.** B and C are *iterative*: from outer iteration 1 each assignment starts from an already-balanced partition. An iterated Lloyd loop with the *existing* solver runs 23 consecutive iterations under the bar. Sinkhorn, implemented and measured, *loses* to the incumbent on 5 of 6 score matrices. Two knobs dismissed as dead — budget and step size — each fix it on their own; the “a $10\times$ budget costs 20 minutes” figure was operations *counted*, never timed (the true cost is 33.9 ms/iteration).

5 The gate, and whether it discriminates

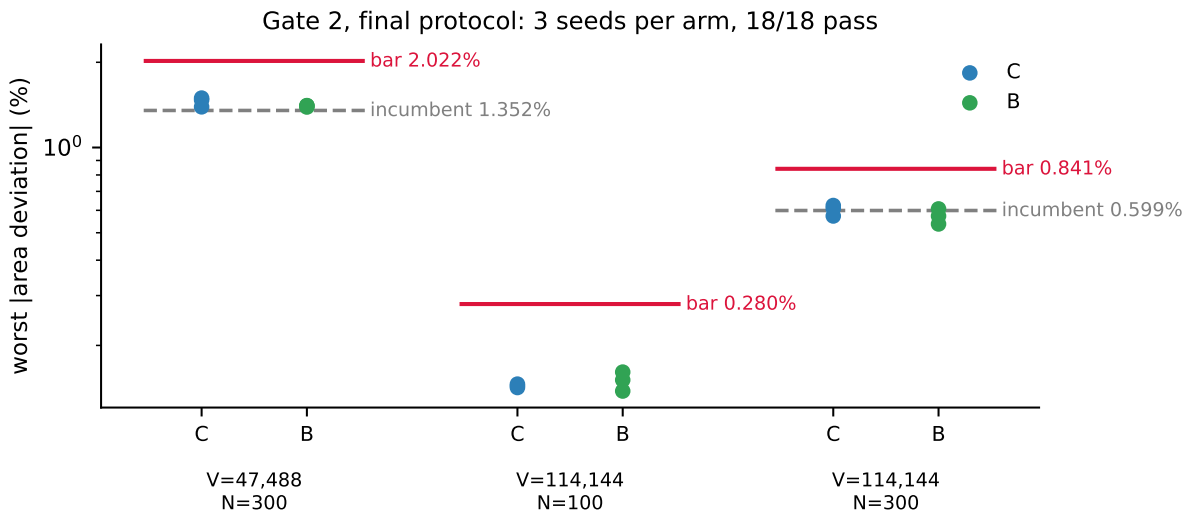


Figure 1: Final protocol: three fixtures, two arms, three seeds, 18/18 pass. Red line: the acceptance bar. Grey dashed: the incumbent at equal budget.

The bar is $\max(2 \times \text{vertex granularity}, \text{A’s own dual stall at equal budget})$. A bar the shipped solver itself fails cannot be evidence about a foreign score scale; a bar below the one-vertex granularity floor is impossible by construction.

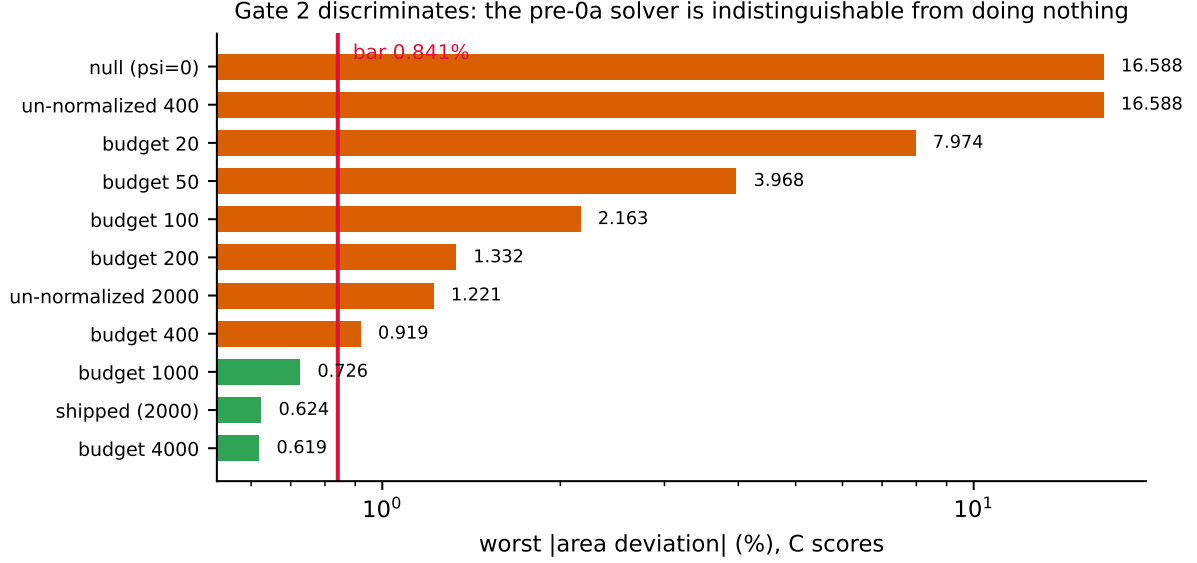


Figure 2: Degraded solvers at the production pin, C scores. The pre-0a un-normalized solver scores 16.588% — *equal to the null solver to three decimals*, because its best iterate is its $\psi=0$ starting point.

The shipped configuration sits 27–48% under its bars with the failure boundary a factor 2–5 away. Gate 3 asserts a null solver *fails*; gate 4 verifies that early stopping is safe inside an iterated loop (3.0× faster for +0.008 points of degradation).

6 The result, stated at equal footing

fixture	bar	C (s0/s1/s2)	B (s0/s1/s2)
V=47,488, N=300	2.022%	1.472 / 1.392 / 1.495	1.405 / 1.390 / 1.406
V=114,144, N=100	0.280%	0.146 / 0.142 / 0.144	0.161 / 0.151 / 0.138
V=114,144, N=300	0.841%	0.624 / 0.573 / 0.609	0.537 / 0.573 / 0.607

Against the incumbent’s 0.5991% at the production pin, C spans 0.573–0.624% and B spans 0.537–0.607%: **parity**, $\pm 10\%$. Earlier drafts of this claim said “2–3× better” and then “clear of parity by 30%”; both compared arms at 2000 iterations against A at 400. *A budget difference wearing a method difference’s clothes*. The assignment step is not a differentiator and was never meant to be one — 0a’s job was to make the shared primitive usable so that perimeter can be measured.

7 Five measurement artefacts read as method results

1. **The stall guard** warned on any floored line search — which is how a healthy level normally ends. Two commits reasoned from the false positive.
2. **A corpus glob** reached 130 of 350 traces, and the resulting “empty gap” became a report’s strongest argument.
3. **A baseline audit** enumerated one working directory and concluded no $N=300$ Phase 2 baseline existed. Two did, in a sibling worktree.

4. **The theatre bar:** gate 2’s bar included $1.25\times$ a strong reference computed *with the solver under test*, so a broken solver inflated its own bar. Six crippled solvers passed, the weakest doing nothing at all.
5. **The early-stop mask:** with early stopping inside the measurement path, gate 2 passed 12/12 while every number hugged its bar from below (Figure 3).



Figure 3: Artefact 5. A passing gate looks identical either way; only the pre-registered prediction distinguished them.

Two things caught these, and neither was insight. **Pre-registration:** the early-stop mask surfaced only because predicted values were committed before the run, and four “missed” in a suspiciously uniform direction. **Asking whether anything fails:** every test asked “*does the shipped solver pass?*” and none asked “*does anything fail?*”, so a gate that passed everything looked exactly like a gate that worked. Gate 3 now closes that.

On “survived adversarial review”. Four adversarial reviews inform this work and each found real defects. They were subagents dispatched *from the same session as the work they reviewed* — independent of the author’s reasoning, not of the author’s framing, tooling, or choice of what to point them at. Every load-bearing number they produced has since been self-reproduced, except one marked indicative in the plan. That caveat belongs next to the claim.

8 Conclusions

1. The harness is qualified: it returns a known answer bitwise, and hardening an arm’s labels costs nothing measurable.
2. No new assignment solver is needed. Normalization, step size and budget suffice; the auction/Sinkhorn conclusion was a fixture artefact.
3. The gate discriminates: 18/18 for the shipped configuration, with the pre-0a solver indistinguishable from doing nothing.
4. Arms and incumbent are at parity $\pm 10\%$ on assignment quality. The open question — whether B or C produces *shorter perimeter* — is untouched by this report.

Related documents

- Plan and pre-registration: docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md
- Taxonomy: docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md
- The gap being closed: docs/reference/winner_take_all_partition_gap.md
- Prior arms: docs/experiments/05-soft-area-constraint/, docs/experiments/06-subfloor-ladder/

References

- [1] Franz Aurenhammer, Friedrich Hoffmann, and Boris Aronov. Minkowski-type theorems and least-squares clustering. *Algorithmica*, 20(1):61–76, 1998.
- [2] Dimitri P. Bertsekas. The auction algorithm: A distributed relaxation method for the assignment problem. *Annals of Operations Research*, 14(1):105–123, 1988.
- [3] Beniamin Bogosel and Édouard Oudet. Partitions of minimal length on manifolds. *Experimental Mathematics*, 26(4):496–508, 2017. YEAR CORRECTED 2026-08-21: this entry read year 2023 with no volume or pages, and its key was bogosel2023partitions. Crossref gives print publication 2 October 2017 (online 4 October 2016), *Experimental Mathematics* 26(4):496–508. This is the foundational reference for the Phase 1 relaxation and Phase 2 refinement methodology.
- [4] Keenan Crane, Clarisse Weischedel, and Max Wardetzky. Geodesics in heat: A new approach to computing distance based on heat flow. *ACM Transactions on Graphics*, 32(5):152:1–152:11, 2013.
- [5] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, pages 2292–2300, 2013.
- [6] Fernando de Goes, Katherine Breeden, Victor Ostromoukhov, and Mathieu Desbrun. Blue noise through optimal transport. *ACM Transactions on Graphics*, 31(6):171:1–171:11, 2012.
- [7] Selim Esedoğlu and Felix Otto. Threshold dynamics for networks with arbitrary surface tensions. *Communications on Pure and Applied Mathematics*, 68(5):808–864, 2015.
- [8] Allen Gersho. Asymptotically optimal block quantization. *IEEE Transactions on Information Theory*, 25(4):373–380, 1979.
- [9] Thomas C. Hales. The honeycomb conjecture. *Discrete & Computational Geometry*, 25(1):1–22, 2001.
- [10] Matt Jacobs, Ekaterina Merkurjev, and Selim Esedoğlu. Auction dynamics: A volume constrained MBO scheme. *Journal of Computational Physics*, 354:288–310, 2018. AUTHORS CORRECTED 2026-08-21, verified against dblp (journals/jcphy/JacobsME18) and ScienceDirect. This entry previously read “Jacobs, Kim and Léger”, an error that originated in the archived A–E proposal (docs/reference/phase1_highn_proposal_ABCDE.original.md) and propagated into seven documents before being checked. Volume, pages and year were correct throughout. Note Esedoğlu is an author of both this and esedoglu2015threshold.
- [11] Jun Kitagawa, Quentin Mérigot, and Boris Thibert. Convergence of a Newton algorithm for semi-discrete optimal transport. *Journal of the European Mathematical Society*, 21(9):2603–2651, 2019.

- [12] Stuart P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [13] Quentin Mérigot. A multiscale approach to optimal transport. *Computer Graphics Forum*, 30(5):1583–1592, 2011.
- [14] Barry Merriman, James K. Bence, and Stanley J. Osher. Motion of multiple junctions: A level set approach. *Journal of Computational Physics*, 112(2):334–363, 1994.
- [15] Luciano Modica and Stefano Mortola. Un esempio di γ -convergenza. *Bollettino dell’Unione Matematica Italiana B (5)*, 14:285–299, 1977. The original Γ -convergence example behind the Phase 1 relaxation energy.
- [16] Yves van Gennip, Nestor Guillen, Braxton Osting, and Andrea L. Bertozzi. Mean curvature, threshold dynamics, and phase field theory on finite graphs. *Milan Journal of Mathematics*, 82(1):3–65, 2014.